SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I S.ROHINI / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.ROHINI

Annexure A

Short Answer Test

1. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

Introduction:

A Multilayer Perceptron (MLP) is a feed-forward Artificial Neural Network (ANN) consisting of an Input Layer, one or more Hidden Layers, and an Output Layer. It is widely used for binary classification problems such as spam detection, disease prediction, and fraud detection.

The Backpropagation Algorithm is a supervised learning algorithm that trains the neural network by minimizing the prediction error. It updates the weights and biases iteratively until the model achieves high accuracy.

Architecture of Multilayer Perceptron:

Multilayer Perceptron (MLP)

Input Layer
|
v
Hidden Layer
|
v
Output Layer
|
v
Output (0 or 1)

Working of Backpropagation:

Backpropagation consists of three main stages.

1. Forward Pass

The input data is given to the input layer. Every neuron calculates the weighted sum of inputs and passes the result through an activation function.

Formula

Net Input

Net = (W1 × X1) + (W2 × X2) + b

Where

- W = Weight
- X = Input
- b = Bias

Example

Input Values

X1 = 1

X2 = 0

Weights

W1 = 0.5

W2 = 0.4

Bias = 0.2

Calculation

Net = (1 × 0.5) + (0 × 0.4) + 0.2

Net = 0.7

Apply the Sigmoid activation function.

Output = 1 / (1 + e^(-Net))

Output = 1 / (1 + e^(-0.7))

Output ≈ 0.67

Thus, the predicted output of the network is 0.67.

2. Error Calculation

The predicted output is compared with the target output.

Formula

Error = Target − Predicted Output

Example

Target Output = 1

Predicted Output = 0.67

Error = 1 − 0.67

Error = 0.33

The objective of training is to reduce this error.

3. Backward Propagation

The calculated error is propagated backward through the network, and all weights are updated.

Weight Update Formula

New Weight = Old Weight + (Learning Rate × Error × Input)

Example

Old Weight = 0.50

Learning Rate = 0.10

Input = 1

Error = 0.33

Calculation

New Weight = 0.50 + (0.10 × 0.33 × 1)

New Weight = 0.533

The updated weight is used in the next iteration.

This process continues until the prediction error becomes minimum.

Role of Learning Rate

The Learning Rate (η) determines how much the weights are updated during each iteration.

- Small learning rate (0.01) → Slow but stable convergence.
- Large learning rate (0.9) → Faster learning but may overshoot the optimal solution.
- Appropriate learning rate → Faster convergence and better accuracy.

Role of Activation Functions

Activation functions introduce non-linearity into the neural network, enabling it to solve complex problems.

1. Sigmoid Function

Output = 1 / (1 + e^(-x))

- Output range: 0 to 1
- Suitable for binary classification.

2. ReLU (Rectified Linear Unit)

ReLU(x) = max(0, x)

- Faster training.
- Reduces vanishing gradient problems.

3. Tanh Function

- Output range: −1 to +1
- Provides better convergence than Sigmoid in many hidden layers.

Applications

- Credit Card Fraud Detection
- Speech Recognition

Advantages

- Minimizes prediction error.
- Improves classification accuracy.

Conclusion:

The Backpropagation Algorithm is the most widely used learning algorithm for training a Multilayer Perceptron (MLP). It performs a forward pass to generate predictions, computes the prediction error, and updates the weights using backward propagation. The learning rate controls the speed of learning, while activation functions improve the model's ability to learn complex patterns. By repeatedly updating the weights, the network converges to an optimal solution and achieves high prediction accuracy.

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2. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models. (5 Marks)

Decision Tree Learning System

Introduction

A **Decision Tree** is a supervised machine learning algorithm used for **classification and regression**. It represents decisions in the form of a tree, where each internal node represents a feature, each branch represents a decision, and each leaf node represents the final prediction.

Architecture of Decision Tree

```
              Decision Tree
                 Outlook
               /   |    \
         Sunny Overcast  Rain
              /           \
        Humidity           Wind
         /    \           /   \
    High   Normal     Strong   Weak
     No      Yes        No      Yes
```

Components of a Decision Tree

1. Root Node

- The first node of the tree.
- Represents the most important feature selected using Information Gain.

2. Decision Node

- Represents a test on a feature.
- Splits the dataset into smaller subsets.

3. Leaf Node

- Represents the final prediction or class label.
- No further splitting occurs.

Feature Selection using Entropy and Information Gain

Decision Trees select the best feature using **Entropy** and **Information Gain**.

Entropy

Entropy measures the impurity or randomness in the dataset.

Formula

$Entropy(S) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$

Where:

- p_1 = Probability of Yes
- p_2 = Probability of No

Interpretation:

- Entropy = 0 → Pure dataset.
- Higher entropy → More randomness.

Information Gain

Information Gain measures the reduction in entropy after splitting the data.

Formula

Information Gain = Entropy(Parent) − Entropy(Children)

The feature with the **highest Information Gain** is selected as the root node.

Outlook	**Humidity**	**Play Tennis**
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Rain	High	No
Rain	Normal	Yes

Tree Construction

Step 1

Calculate the entropy of the complete dataset.

Step 2

Calculate the Information Gain for each feature:

- Outlook
- Humidity

Step 3

Select the feature with the **highest Information Gain** as the root node.

Step 4

Split the dataset into subsets.

Step 5

Repeat the process until all leaf nodes contain a single class.

The final tree predicts whether the person will **Play Tennis (Yes/No)**.

Pruning

Pruning removes unnecessary branches from the decision tree.

Types of Pruning

1. Pre-Pruning

- Stops tree growth early.
- Prevents unnecessary splits.

2. Post-Pruning

- Builds the complete tree first.
- Removes branches that do not improve accuracy.

Advantages of Pruning

- Reduces tree size.
- Improves prediction accuracy.

Overfitting in Decision Trees

Overfitting occurs when the model memorizes the training data instead of learning general patterns.

Methods to Control Overfitting

- Pruning
- Limit Maximum Tree Depth
- Increase Minimum Samples per Leaf
- Use Cross Validation
- Use Ensemble Methods (Random Forest)

These techniques improve the model's ability to predict unseen data.

Applications

- Medical Diagnosis

- Loan Approval

Advantages

- Easy to understand and interpret.
- Handles both numerical and categorical data.

Conclusion

A **Decision Tree** is a simple and effective supervised learning algorithm that classifies data using a tree-like structure. It selects the best features using **Entropy** and **Information Gain**, constructs the tree by recursively splitting the dataset, and uses **pruning** to reduce overfitting. Due to its simplicity, interpretability, and high accuracy, Decision Trees are widely used in real-world machine learning applications.

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3. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

Convolutional Neural Network (CNN)

Introduction

A **Convolutional Neural Network (CNN)** is a type of **Deep Learning** model mainly used for **image classification, object detection, facial recognition, and medical image analysis**. CNN automatically extracts important features from images using convolution and pooling operations.

```
Input Image
     |
     v
Convolution Layer
     |
     v
ReLU Layer
     |
     v
Pooling Layer
     |
     v
```

Fully Connected Layer

|

v

Output Class

Layers of CNN

1. Input Layer

- Receives the input image.
- Stores pixel values of the image.
- Example: 28 × 28 grayscale image.

2. Convolution Layer

- Uses filters (kernels) to extract important features.
- Detects edges, corners, textures, and shapes.
- Produces **Feature Maps**.

Formula

Feature Map = Input Image * Filter

(* represents convolution operation.)

3. ReLU (Activation Layer)

ReLU introduces non-linearity into the model.

Formula

ReLU(x) = max(0, x)

Advantages

- Faster training
- Reduces vanishing gradient problem
- Improves learning efficiency

4. Pooling Layer

Pooling reduces the size of the feature map while preserving important information.

Types

- Max Pooling
- Average Pooling

Example:

Original Feature Map

6 4

8 5

Max Pooling Output

Advantages

- Reduces computation
- Prevents overfitting
- Speeds up training

5. Fully Connected Layer

- Connects all neurons.
- Combines extracted features.
- Performs final classification.

Example:

Cat Image → Cat

Dog Image → Dog

Feature Map Generation

Feature maps are generated by sliding a filter over the input image.

Example

Input Image

1 2 1

0 1 2

2 1 0

Filter

1 0

0 1

The filter moves across the image and extracts important features like edges and textures.

These outputs form the **Feature Map**.

Backpropagation in CNN

Backpropagation updates the filter weights to minimize prediction error.

Step 1

Forward pass generates prediction.

Step 2

Calculate error.

Error = Target − Predicted Output

Step 3

Update weights.

New Weight = Old Weight + (Learning Rate × Error × Input)

The process repeats until the error becomes minimum.

Image Classification Example

Problem: Identify whether the image is a Cat or Dog.

Steps

1. Input image is provided.
2. Convolution layer extracts image features.
3. ReLU introduces non-linearity.
4. Pooling reduces image size.
5. Fully connected layer classifies the image.
6. Output:

Prediction = Cat

Factors Affecting CNN Performance

- Quality of training dataset
- Number of training images
- Filter (Kernel) size
- Learning rate
- Number of epochs
- Activation function
- Batch size
- Overfitting and underfitting

Applications

- Face Recognition
- Traffic Sign Detection

Advantages

- Automatic feature extraction
- Suitable for large image datasets

Conclusion

A **Convolutional Neural Network (CNN)** is a powerful deep learning model designed for image-related tasks. It consists of **Convolution, ReLU, Pooling, and Fully Connected layers**, which work together to extract features and classify images accurately. **Backpropagation** updates the filter weights iteratively, while proper selection of **learning rate, filter size, and training data** improves model performance. CNNs are widely used in computer vision applications due to their high accuracy and efficient feature extraction.

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4. With the help of a neat diagram, explain the architecture of an **Artificial Neural Network (ANN)**. Describe the roles of input, hidden, and output layers, and explain how weights and biases are used in computation. Illustrate how forward propagation works with a simple example, and discuss how overfitting can be controlled using techniques such as dropout and regularization. (5 Marks)

Artificial Neural Network (ANN) Architecture

Introduction

An **Artificial Neural Network (ANN)** is a supervised machine learning model inspired by the human brain. It consists of interconnected neurons organized into **Input Layer, Hidden Layer, and Output Layer**. ANN is widely used for **classification, regression, image recognition, speech recognition, and disease prediction**.

Architecture of ANN

Input Layer

|

v

Hidden Layer

|

v

Output Layer

|

v

Final Prediction

Layers of ANN

1. Input Layer

- First layer of the network.
- Receives input data.
- Each neuron represents one input feature.

Example

Patient Data:

- Age
- Blood Pressure
- Glucose Level

These values are passed to the hidden layer.

2. Hidden Layer

- Performs mathematical computations.
- Learns patterns from the data.
- Can contain one or more hidden layers.

The hidden layer applies weights, bias, and activation functions to generate meaningful outputs.

3. Output Layer

- Produces the final prediction.
- Number of neurons depends on the problem.

Example

Disease Prediction

Output = Yes

or

Output = No

Weights and Biases

Weights

Weights determine the importance of each input feature.

Higher weight means greater influence on the output.

Bias

Bias shifts the activation function and improves learning.

Without bias, the model may not fit the data accurately.

Computation in ANN

Each neuron computes the weighted sum of inputs.

Formula

$Net = (W1 \times X1) + (W2 \times X2) + b$

Where

- W = Weight
- X = Input
- b = Bias

Forward Propagation

Forward propagation is the process of passing input data through the network to generate the output.

Step 1

Input values

$X1 = 1$

$X2 = 0$

Weights

W1 = 0.5

W2 = 0.4

Bias = 0.2

Step 2

Calculate Net Input

Net = (1 × 0.5) + (0 × 0.4) + 0.2

Net = 0.7

Step 3

Apply Sigmoid Activation Function

Output = 1 / (1 + e^(-Net))

Output = 1 / (1 + e^(-0.7))

Output ≈ 0.67

The network predicts an output of **0.67**.

Activation Functions

1. Sigmoid

Output = 1 / (1 + e^(-x))

- Output between 0 and 1.
- Used for binary classification.

2. ReLU

ReLU(x) = max(0, x)

- Faster training.
- Eliminates negative values.

3. Tanh

- Output ranges from **−1 to +1**.
- Often used in hidden layers.

Overfitting

Overfitting occurs when the ANN learns the training data too well and performs poorly on new data.

Techniques to Control Overfitting

1. Dropout

- Randomly disables some neurons during training.
- Prevents dependence on a few neurons.
- Improves model generalization.

2. Regularization

Regularization adds a penalty to large weights.

Types

- L1 Regularization
- L2 Regularization

Benefits:

- Reduces model complexity.
- Prevents overfitting.
- Improves prediction accuracy.

3. Early Stopping

Stops training when validation accuracy stops improving.

4. Cross Validation

Evaluates the model on different subsets of the dataset to improve reliability.

Applications

- Image Recognition
- Face Recognition

Advantages

- Learns complex patterns automatically.
- Improves performance with more training data.

Conclusion

An **Artificial Neural Network (ANN)** consists of **Input, Hidden, and Output layers** that work together to learn patterns from data. During **forward propagation**, the network computes weighted sums, adds bias, and applies activation functions to produce predictions. **Weights and biases** improve learning, while **dropout, regularization, early stopping, and cross-validation** help control overfitting and improve the model's accuracy and generalization.

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5. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently. (5 Marks)

Genetic Algorithm for Optimization

Introduction

A **Genetic Algorithm (GA)** is an optimization and search technique inspired by the process of **natural selection**. It is used to find the best solution for complex optimization problems such as **task scheduling, route optimization, feature selection, and resource allocation**. GA evolves a population of candidate solutions over several generations to obtain the optimal or near-optimal solution.

Working of Genetic Algorithm:

Initial Population
|
v
Fitness Evaluation
|
v
Selection
|
v
Crossover
|
v
Mutation
|
v
New Population
|
v
Best Optimized Solution

Chromosome Representation

In Genetic Algorithms, each possible solution is called a **Chromosome**.

Each chromosome consists of **Genes**, where every gene represents one part of the solution.

Example

Task Scheduling

Chromosome	**Task Order**
C1	A B C D
C2	B A D C
C3	C D A B
C4	D B A C

Each chromosome represents a different schedule.

Fitness Function

The **Fitness Function** measures the quality of each chromosome.

A chromosome with a **higher fitness value** is considered a better solution.

Example

Chromosome	Fitness Value
C1	60
C2	75
C3	90
C4	65

Here, **C3** is the best chromosome because it has the highest fitness value.

Selection

Selection chooses the best chromosomes from the population to become parents for the next generation.

Methods of Selection

- Roulette Wheel Selection
- Tournament Selection
- Rank Selection

Example

Population

C1 → 60

C2 → 75

C3 → 90

C4 → 65

Selected Parents

C2 and C3

These chromosomes have higher fitness values and are more likely to produce better offspring.

Crossover

Crossover combines two parent chromosomes to create new offspring.

Example

Parent 1

A B | C D

Parent 2

D C | A B

After Single-Point Crossover

Offspring 1

A B A B

Offspring 2

D C C D

This operation combines the strengths of both parents.

Mutation

Mutation randomly changes one or more genes in a chromosome.

Example

Before Mutation

A B A B

After Mutation

A D A B

Mutation introduces diversity into the population and prevents the algorithm from getting stuck in a local optimum.

Evolution Process

Generation 1

Chromosome	**Fitness**
C1	60
C2	75
C3	90
C4	65

Best Chromosome = **C3**

↓

Apply Selection, Crossover, and Mutation

↓

Chromosome	**Fitness**
O1	92
O2	95
O3	90
O4	88

Best Chromosome = **O2**

The fitness value improves from **90 to 95**, indicating a better optimized solution.

Advantages of Genetic Algorithm

- Finds near-optimal solutions efficiently.
- Handles large and complex search spaces.

Applications

- Task Scheduling
- Feature Selection

Conclusion

A **Genetic Algorithm** is a powerful optimization technique that represents solutions as **chromosomes** and evaluates them using a **fitness function**. Through **selection, crossover, and mutation**, better solutions are generated in each generation. These evolutionary operations improve the fitness of the population and help identify a **near-optimal solution** efficiently for complex optimization problems such as scheduling, feature selection, and routing.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand